# PROJECT PROPOSAL

B.Tech Computer Science Engineering (Artificial Intelligence & Machine Learning)
SRM University AP • Academic Year 2024–25

# Smart Vehicle Intelligence and Enforcement System (SVIES)

Real-Time AI-Powered Multi-Modal Vehicle Monitoring with Fake Plate Detection, Automated Authority Notification, and Offline Edge Intelligence

## GROUP DETAILS


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## 1. Title of the Project

**Smart Vehicle Intelligence and Enforcement System (SVIES): Real-Time AI-Powered Multi-Modal Vehicle Monitoring with Fake Plate Detection, Helmet/Seatbelt Enforcement, Geofencing Intelligence, Repeated Offender Tracking, and Offline Edge Processing**

## 2. Abstract

Existing Automatic Number Plate Recognition (ANPR) systems focus narrowly on plate detection and character recognition, offering little beyond basic identification. This proposal presents the Smart Vehicle Intelligence and Enforcement System (SVIES) — an end-to-end, context-aware, AI-powered vehicle monitoring platform that goes far beyond simple detection.

SVIES integrates YOLOv8-based license plate and vehicle detection with a seven-layer intelligence engine that cross-references live government transport databases (VAHAN/Parivahan), Pollution Under Control Certificate (PUCC) records, police stolen vehicle databases, and insurance registries in real time. A novel Fake and Mismatched Plate Detection module identifies forged plates through vehicle-type mismatch analysis, color-code validation, font anomaly detection, and dual-location duplicate alerts — capabilities absent in all existing Indian ANPR deployments.

The system further integrates helmet and seatbelt violation detection using YOLOv8-pose, blacklist zone geofencing with priority alerting, repeated offender intelligence scoring, and an offline edge-processing mode for low-connectivity deployments. Upon flagging a vehicle, the system automatically notifies relevant authorities (police, RTO, traffic enforcement) and the registered vehicle owner via SMS, email, and WhatsApp, complete with GPS location, CCTV snapshot, and a cryptographically timestamped violation summary.

This system addresses critical real-world enforcement gaps on Indian roads and demonstrates genuine novelty in fake plate forensics, multi-modal violation detection, database integration, and real-time decision intelligence — making it suitable for real-world smart city deployment as well as research publication.

## 3. Existing System and Its Drawbacks

### 3.1 Existing Systems

Current Automated License Plate Recognition (ALPR) and vehicle monitoring systems employed across India and globally include commercial solutions such as Hi-Tech ANPR cameras (deployed at toll plazas), Genetec AutoVu (used at airports and border checkposts), and open-source implementations based on OpenALPR. Academic systems typically combine Tesseract OCR with basic edge detection. While these systems are useful for isolated plate reading, they are fundamentally limited to text extraction and do not provide contextual vehicle intelligence.

### 3.2 Summary of Drawbacks

| Limitation | Impact on Enforcement |
|---|---|
| No fake or mismatched plate detection | Criminals exploit vehicle-plate type mismatches freely |
| Limited OCR accuracy in Indian conditions (rain, dust, night) | High false-negative rate on real roads |
| No support for BH-series and regional language plates | Modern Indian plates go unrecognized |
| No cross-referencing with VAHAN, PUCC, insurance, stolen vehicle databases | Non-compliant vehicles pass without detection |
| No automated alert mechanism for authorities or owners | Violations go unreported in real time |
| No helmet or seatbelt violation detection | Traffic safety enforcement requires manual presence |
| No geofencing or zone-based priority enforcement | High-risk zones treated same as regular roads |
| No repeated offender tracking or escalation logic | Habitual violators treated identically each time |
| No offline or edge-processing capability | Rural and low-bandwidth checkposts cannot be served |
| High cost — commercial systems are unaffordable for small municipalities | Enforcement gaps in Tier 2/3 cities and rural areas |

## 4. Proposed System and Its Advantages

### 4.1 System Overview

SVIES is a seven-layer intelligent pipeline that transforms a live camera feed into actionable enforcement intelligence. Each layer operates in sequence, with outputs feeding the next, and all layers collectively contributing to a unified risk profile per detected vehicle.

#### Layer 1 — Video Capture & Preprocessing

Live CCTV or IP camera feed (minimum 1080p) with real-time frame extraction at 25–30 FPS. Includes adaptive preprocessing: automatic brightness/contrast normalization for night and adverse weather conditions, motion-blur reduction via frame differencing, and ROI (Region of Interest) masking to focus compute resources on lane-level vehicle zones.

#### Layer 2 — AI Detection Engine

YOLOv8 (fine-tuned on the IDD — Indian Driving Dataset and custom Indian plate dataset) simultaneously detects: (a) license plate bounding boxes, (b) vehicle type classification —

2-wheeler, 3-wheeler, car, SUV, bus, truck, (c) vehicle color via HSV histogram analysis, and (d) estimated manufacturing year range using a ResNet50 age classifier trained on vehicle silhouette features.

### Layer 2.5 — Fake & Mismatched Plate Detection Module

This module is the primary differentiating contribution of SVIES. It performs four independent forensic checks on each detected plate:

- Vehicle-Type vs. Plate Format Mismatch: India assigns distinct plate series to vehicle categories. A scooty (2-wheeler) bears a different registration series than a car. YOLOv8 classifies the vehicle type; the OCR-extracted plate number is then cross-checked against VAHAN to confirm the registered vehicle category. If a car carries a plate registered to a 2-wheeler, the system raises a FAKE_PLATE_TYPE alert.
- Plate Color Code Violation: Under CMVR (Central Motor Vehicles Rules), plate colors are legally prescribed — white background for private vehicles, yellow for commercial, green for EVs, black with yellow text for rentals. A computer vision color classifier checks the detected plate background color against the vehicle classification. Mismatches trigger a PLATE_COLOR_MISMATCH flag.
- IS 10731 Font & Format Anomaly Detection: Genuine Indian plates conform to IS 10731 specifications for character height, spacing, and font. A trained CNN classifier evaluates character morphology and inter-character spacing. Non-standard fonts, DIY plates, and handwritten characters are flagged as PLATE_FORMAT_ANOMALY.
- Duplicate Plate / Multi-Location Alert: When the same plate number is detected at two geographically separated cameras simultaneously (within a configurable time window), the system raises a DUPLICATE_PLATE alert, indicating one or both vehicles are using cloned plates.
- State Code vs. Registration State Mismatch: If a plate bears a state prefix (e.g., MH for Maharashtra) but VAHAN confirms the vehicle is registered in a different state (e.g., Telangana), a STATE_MISMATCH flag is raised — a common fraud tactic to evade local traffic enforcement.

### Layer 3 — OCR and Plate Parsing

EasyOCR combined with Tesseract (ensemble approach for higher accuracy) with post-processing rules for all Indian plate formats: standard, BH-series (Bharat series), and regional language plates (Tamil, Telugu, Malayalam scripts). Confidence scoring resolves ambiguous characters (0/O, 1/I, 8/B) using contextual plate-format grammar.

### Layer 4 — Multi-Database Intelligence & Risk Scoring

Real-time API calls are made to (i) VAHAN — vehicle registration and ownership, (ii) PUCC database — pollution certificate validity, (iii) Insurance registry — third-party and comprehensive insurance status, and (iv) Police stolen vehicle database — national and state-level theft records.

A dynamic Risk Score is computed per vehicle:

| Violation Parameter | Weight | Alert Level |
|---|---|---|
| Stolen vehicle | 40 | CRITICAL |
| Fake / mismatched plate detected | 35 | CRITICAL |

| Violation Parameter | Weight | Alert Level |
|---|---|---|
| No valid registration | 25 | HIGH |
| Expired insurance | 20 | HIGH |
| No valid PUCC | 15 | MEDIUM |
| Helmet / seatbelt violation | 10 | MEDIUM |
| Blacklist zone entry | 15 | HIGH |
| Repeated offender (3+ violations in 30 days) | 20 | HIGH |

### Layer 4.5 — Helmet & Seatbelt Violation Detection

Using YOLOv8-pose keypoint estimation, the system detects driver and passenger body positions. A binary helmet classifier (EfficientNet-B0, fine-tuned on Indian road imagery) determines helmet presence on 2-wheeler riders. For 4-wheelers, a seatbelt classifier analyzes the driver-side ROI. Violations are logged with the vehicle plate number and automatically added to the vehicle's violation record, contributing to the Risk Score.

### Layer 5 — Blacklist Zone Geofencing & Automated Enforcement Alerts

A geofencing engine (OpenStreetMap + custom zone polygons) defines priority enforcement zones: school zones, hospital zones, government buildings, residential areas, and low-emission corridors. When a flagged vehicle enters a designated blacklist zone, the alert priority is automatically escalated. Zone types and their alert routing:

- School/Hospital Zone — Stolen or fake-plate vehicle: Immediate dispatch alert to nearest police control room + school/hospital security
- Low-Emission Zone — High-pollution vehicle (expired PUCC, older vehicle): RTO notification + zone denial log
- Government Security Zone — Blacklisted vehicle: Dual alert to police and security agency

All alerts are delivered via SMS (Twilio), Email (SMTP), and optionally WhatsApp Business API, and are cryptographically timestamped using SHA-256 hashing to ensure legal admissibility.

### Layer 5.5 — Repeated Offender Intelligence

SVIES maintains a persistent violation history database (SQLite/PostgreSQL) for every detected vehicle plate. An offender scoring engine escalates the alert level when a vehicle accumulates violations:

- 1–2 violations in 30 days: Standard alert to owner and local RTO
- 3–5 violations in 30 days: Escalation to district traffic superintendent + owner notification with court challan warning
- 6+ violations in 30 days: Red-flag status — alerts to state traffic authority, auto-generation of court summons template, and inclusion in high-priority watch list

### Layer 6 — Offline Edge Processing Mode

For deployment at rural checkposts and low-bandwidth areas, SVIES includes a lightweight edge mode: YOLOv5n (nano model, 1.9M parameters) runs on a Raspberry Pi 4 or any i5-class CPU. A local SQLite database caches the most recent 30-day vehicle records from VAHAN/police databases. Violations detected offline are queued and synced to the central server when connectivity resumes. This makes SVIES the first Indian ANPR-class system designed for zero-internet deployment.

### Layer 7 — Dashboard and Reporting

A Streamlit/React dashboard provides real-time monitoring with GPS heat maps of violation clusters, vehicle timeline queries, repeat offender leaderboards, and auto-generated monthly PDF reports for submission to the RTO and traffic authority. Historical data exports are available in CSV and JSON formats.

*Note: Due to restricted access to live government APIs, the prototype uses simulated (mock) databases that replicate real-world VAHAN, PUCC, insurance, and police systems, preserving identical system logic and enforcement workflows.*

## 4.2 Advantages of SVIES

| Feature | SVIES Advantage |
|---|---|
| **Fake plate detection** | 5-mode forensic analysis — unique in Indian ANPR research |
| **YOLOv8 detection** | State-of-the-art real-time object detection, outperforms YOLOv5/v7 |
| **Night & adverse weather support** | Adaptive preprocessing ensures consistent accuracy |
| **BH-series & regional language plates** | Full coverage of all modern Indian plate formats |
| **Helmet & seatbelt detection** | Integrated safety enforcement without separate hardware |
| **Geofencing with zone-priority alerts** | Context-aware enforcement, not generic flagging |
| **Repeated offender intelligence** | Tiered escalation — first system to implement this in Indian context |
| **Offline edge processing** | Rural and remote deployment capability |
| **Zero production cost** | 100% open-source stack: YOLOv8, EasyOCR, Python, Streamlit |
| **Cryptographic timestamping** | Alerts are court-admissible evidence |

## 5. Block Diagram of the Proposed System

The SVIES processing pipeline is structured as follows:

**[ CAMERA INPUT — Live CCTV / IP Camera / Uploaded Video (1080p) ]**

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LAYER 1: Preprocessing → Frame extraction | Night normalization | ROI masking

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LAYER 2: YOLOv8 Detection → Plate bounding box | Vehicle type | Color | Age estimation

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**LAYER 2.5: FAKE PLATE DETECTION → Type mismatch | Color code | Font anomaly | Duplicate | State mismatch**

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LAYER 3: OCR + Plate Parsing → EasyOCR + Tesseract ensemble | BH-series | Regional scripts

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LAYER 4: Database Intelligence + Risk Scoring → VAHAN | PUCC | Insurance | Stolen vehicles | Risk Score

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**LAYER 4.5: Helmet / Seatbelt Detection → YOLOv8-pose | EfficientNet-B0 classifier**

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**LAYER 5: Geofencing + Automated Alerts → Zone classification | Priority routing | SMS/Email/WhatsApp**

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**LAYER 5.5: Repeated Offender Intelligence → Violation history | Escalation tiers | Court summons**

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**LAYER 6 (EDGE): Offline Mode → YOLOv5n on CPU/Raspberry Pi | Cached DB | Sync-on-connect**

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**LAYER 7: Dashboard & Reporting → Streamlit/React | GPS Heatmaps | Monthly PDF Reports | API Export**

## 6. Hardware and Software Details

### 6.1 Hardware Requirements

| Component | Specification | Purpose |
|---|---|---|
| **Camera** | HD CCTV / IP Camera, min. 1080p @ 30FPS, IR night vision | Primary input; IR for night-mode support |
| **Processing Unit** | Intel i5 / Ryzen 5 or above; 8GB RAM min (16GB recommended) | Core inference and database operations |

| Component | Specification | Purpose |
|---|---|---|
| **GPU (Optional)** | NVIDIA GTX 1060+ / RTX series OR Google Colab T4 | YOLOv8 model training and inference acceleration |
| **Edge Device** | Raspberry Pi 4B (4GB) or equivalent ARM Cortex-A72 | Offline edge-mode deployment |
| **Storage** | 256GB SSD minimum | Video buffer, violation logs, cached database |
| **GPS Module** | NEO-6M GPS (optional) | Mobile deployment unit location tagging |
| **GSM/LTE Module** | SIM800L or 4G LTE dongle | SMS fallback and connectivity in offline areas |
| **Cloud Server** | AWS EC2 t2.medium / Azure B2s (optional) | Central dashboard and database hosting |

## 6.2 Software Requirements

### Frontend

- React.js / Streamlit — Interactive real-time monitoring dashboard
- Leaflet.js / Google Maps API — Geofencing zone visualization and GPS heatmaps
- Bootstrap 5 / Tailwind CSS — Responsive UI styling
- Chart.js / Recharts — Violation analytics and trend visualization

### Backend

- Python 3.10+ — Primary programming language
- FastAPI / Flask — REST API server for frontend-backend communication
- PostgreSQL / SQLite — Violation history, offender records, cached VAHAN data
- Redis — Real-time alert queue and duplicate plate detection cache

### Computer Vision & AI

- YOLOv8 (Ultralytics) — Primary detection: plates, vehicles, helmet, seatbelt
- YOLOv5n — Lightweight edge-mode inference model
- EasyOCR + Tesseract 5.0 — Ensemble OCR with Indian plate post-processing
- OpenCV 4.8 — Frame extraction, preprocessing, color analysis
- ResNet50 / EfficientNet-B0 (PyTorch) — Vehicle age estimation, helmet classification
- Custom CNN — IS 10731 font anomaly detection

### Alert & Communication

- Twilio API — SMS alerts to vehicle owner and authorities (free tier: 1,000 SMS/month)
- SMTP / SendGrid — Email dispatch with CCTV snapshot attachments
- WhatsApp Business API — Officer alert notifications (optional)

- SHA-256 Cryptographic Hashing — Tamper-proof timestamp on all alert records

### Development & Deployment Tools

- GitHub — Version control and collaboration (free)
- Google Colab Pro (free tier) — Model training
- Docker — Containerized deployment
- OpenStreetMap / Nominatim — Free geofencing and geocoding (zero cost)

## 7. Applications and Possible Areas of Implementation

| Application Area | How SVIES Enables It |
|---|---|
| **Fake Plate Enforcement** | Real-time identification of forged, mismatched, or cloned plates — prevents registration fraud and supports crime investigation |
| **Traffic Law Enforcement** | Automated detection of uninsured, unregistered, or pollution-non-compliant vehicles — reduces manual patrolling effort |
| **Stolen Vehicle Recovery** | Instant SMS + email alerts to nearest police station with GPS and CCTV snapshot when a stolen vehicle is detected |
| **Helmet & Seatbelt Enforcement** | Camera-based detection of violations without human presence — applicable at high-accident-rate junctions |
| **Smart City Infrastructure** | Integration with city-wide camera networks for centralised vehicle intelligence and data-driven traffic management |
| **Low-Emission Zone Enforcement** | Identify and flag high-pollution vehicles entering green corridors or eco-sensitive zones |
| **Toll and Entry Gate Automation** | Instant vehicle verification without human operators — reduces congestion and corruption |
| **Campus / Residential Security** | Whitelist/blacklist-based access control; repeated offender tracking within private communities |
| **Rural & Remote Checkpost** | Offline edge mode enables enforcement at locations with no internet connectivity |
| **Research Publication** | Novel fake plate detection methodology and repeated offender intelligence engine are publishable contributions in IEEE/Springer venues |

# 8. Expected Outcomes / Objectives

## 8.1 Primary Objectives

- Develop a fully functional SVIES prototype capable of real-time plate detection, OCR, and multi-database cross-referencing on a standard laptop or edge device
- Implement and validate the five-mode Fake and Mismatched Plate Detection module with measurable precision/recall metrics on a curated test dataset
- Achieve end-to-end alert latency of under 3 seconds from plate detection to SMS dispatch on a standard hardware setup
- Deploy the Repeated Offender Intelligence engine with full three-tier escalation logic and automated court summons template generation
- Demonstrate offline edge-mode operation on a Raspberry Pi 4 or equivalent ARM device with local database syncing

## 8.2 Expected Performance Metrics

| Metric | Target Value |
|---|---|
| Plate detection accuracy (YOLOv8, daytime) | **≥ 95% mAP@0.5** |
| Plate detection accuracy (YOLOv8, night/adverse weather) | **≥ 85% mAP@0.5** |
| OCR accuracy on standard Indian plates | **≥ 92% character-level accuracy** |
| Fake plate detection precision | **≥ 88% (across all 5 check modes)** |
| Helmet detection accuracy | **≥ 90% on 2-wheeler frames** |
| End-to-end alert latency | **< 3 seconds (plate detection → SMS sent)** |
| Offline edge mode FPS (Raspberry Pi 4) | **≥ 10 FPS (YOLOv5n)** |
| Database cross-reference time (mock APIs) | **< 500ms per vehicle** |

## 8.3 Publication and Deployment Goals

- Prepare and submit a research paper on the novel Fake Plate Detection module to an IEEE or Springer conference (ICIP, CVIP, or similar)
- Open-source the complete codebase on GitHub for community validation and research reproducibility
- Deploy a live demo on Streamlit Cloud or Hugging Face Spaces using pre-recorded video for public demonstration at zero cost

## 9. Timeline / Work Plan (10 Weeks)

| Week(s) | Phase | Deliverables |
|---|---|---|
| **1–2** | **Research & Dataset Collection** | Literature review, IDD dataset download, mock DB design, YOLOv8 fine-tuning plan |
| **3** | **Core Detection Module** | YOLOv8 plate + vehicle detector trained and evaluated; preprocessing pipeline ready |
| **4** | **OCR + Plate Parsing** | EasyOCR/Tesseract ensemble with IS 10731 grammar post-processor; BH-series support |
| **5** | **Fake Plate Detection Module** | All 5 forensic checks implemented and tested on synthetic dataset; precision/recall report |
| **6** | **Database Integration + Risk Scoring** | Mock VAHAN/PUCC/insurance/police APIs integrated; risk score engine operational |
| **7** | **Helmet Detection + Geofencing** | YOLOv8-pose helmet classifier; OpenStreetMap zone engine with priority alert routing |
| **8** | **Offender Intelligence + Offline Mode** | 3-tier repeated offender system; YOLOv5n edge mode on Raspberry Pi / CPU-only machine |
| **9** | **Dashboard + Alert System** | Streamlit/React dashboard live; Twilio SMS + email alerts operational; heatmaps functional |
| **10** | **Testing, Documentation & Submission** | Full system integration test, performance benchmarking, research paper draft, final report |

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